MACROECONOMICS

MACRO-11

Time: 5 minutes

Outcome: Calculate output per worker and interpret productivity changes.

Difficulty: Intermediate

Productivity Measurement

THE CORE IDEA

Labor productivity measures output per worker or per hour worked. 2,400 / 300 = 8 units per hour. Productivity = 500/50 = 10 units per hour. Productivity compares output with the labor input used to produce it, so changes in both output and labor matter. A country can consume more when its workers produce more output per worker.

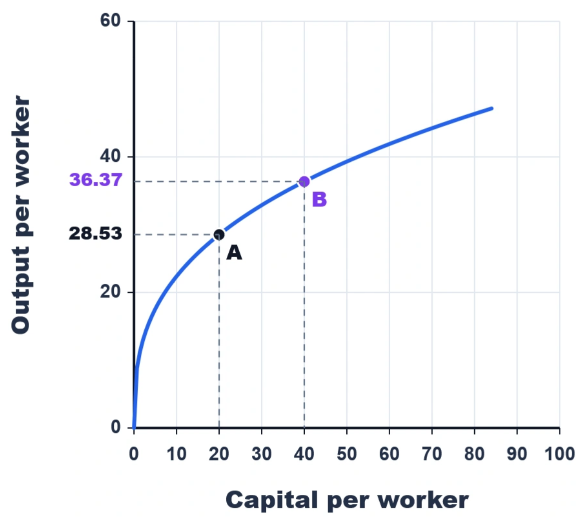
HOW TO RECOGNIZE IT

- Measure labor productivity as real output per worker or per hour worked.
- Compare output using the same amount of labor when evaluating a productivity change.
- An increase in total output is not necessarily a productivity increase if labor input also rises.
- On a production-function graph, diminishing returns make equal additions of capital add progressively less output.

WATCH OUT

Do not confuse total output with output per unit of labor. Productivity compares output with the labor input used to produce it, so changes in both output and labor matter.

WORKED EXAMPLE: MEASURING LABOR PRODUCTIVITY



The graph shows output per worker of 28.53 when capital per worker is 20 and 36.37 when capital per worker is 40. The increase in labor productivity is 36.37 - 28.53 = 7.84 units per worker. Productivity measures output relative to labor input, not total output alone.

CHECK YOURSELF

A business uses more labor hours but produces the same amount of output. What happens to labor productivity?

READY? Return to the game and master this concept.